DIPLOMA PROGRAMME PRACTICE · HL · SET B

Biology / Paper 2

Question booklet · 80 marks · 150 minutes

Show working, units and suitable precision. Use a calculator. Use separate answer paper for additional working and diagrams. Answer all of Section A and two of three Section B questions. Each extended response carries 15 content marks and one mark for overall quality.

SECTION A · COMPULSORY
QUESTION 1 [12]

Testing two guide RNAs

Researchers target a bacterial reporter gene with Cas9 and guide A or B. A control receives Cas9 without a targeting guide. Equal numbers of colonies are sampled after treatment. A sequencing test detects the intended edit and a separate test detects an edit at one predicted off-target locus. The two counts can overlap.

Treatment	Colonies tested	Intended edit	Off-target edit
Control	200	2	1
Guide A	200	156	18
Guide B	200	124	4

(a) Calculate the intended-edit percentage and off-target percentage for each guide. **[3]**

(b) For guide A, calculate the possible range of colonies with both the intended edit and no detected off-target edit. Explain why one exact value cannot be found. **[3]**

(c) Explain the roles of complementarity and Cas9 in the targeting mechanism, and why an imperfect off-target match could be relevant. **[3]**

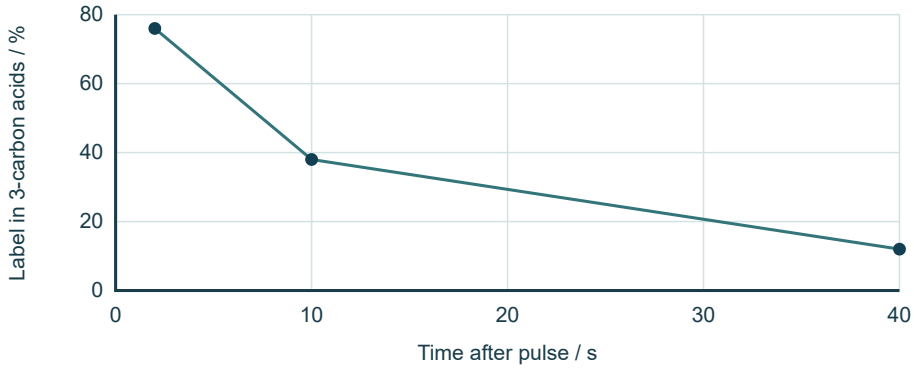
(d) Assess the claim that B is the safer guide for all purposes and propose a stronger test. **[3]**

SECTION A · COMPULSORY
QUESTION 2 [12]

Following labelled carbon in an alga

A $^{14}CO_2$ pulse is delivered to an illuminated culture and then replaced by unlabelled CO_2. The table reports the percentage of all recovered label in three fractions. Recoveries are normalized separately at each time.

Time after pulse / s	3-carbon acids / %	Sugar phosphates / %	Other products / %
2	76	16	8
10	38	43	19
40	12	49	39



Label in 3-carbon acids / % plotted against Time after pulse / s. Points are measured means or observations as tabulated.

(a) Quantify the change in the share of label in 3-carbon acids between 2 s and 40 s in two ways. **[3]**

(b) Explain why early label in 3-carbon acids is consistent with the first fixation step of the Calvin cycle. **[3]**

(c) Explain how light-dependent products enable some of this carbon to reach carbohydrate while fixation continues. **[3]**

(d) Explain why the percentages do not prove that every labelled atom passed directly from the first listed fraction to the second, and suggest an improvement. **[3]**

SECTION A · COMPULSORY
QUESTION 3 [12]

Pressure differences and blood flow

A model systemic circulation has flow F=(arterial pressure−venous pressure)/resistance. Pressures are in kPa. Values below are means over a cardiac cycle. Resting cardiac output is 5.0 dm³ min⁻¹.

Condition	Mean arterial pressure	Mean venous pressure	Relative systemic resistance
Rest	13.0	1.0	1.00
Exercise	15.0	1.5	0.75

(a) Calculate the pressure difference at rest and exercise and find the predicted exercise-to-rest flow ratio. **[3]**

(b) Calculate predicted exercise output and stroke volume if exercise heart rate is 125 min⁻¹. **[3]**

(c) Explain how arterioles in active muscle can contribute to a lower vascular resistance and altered distribution of flow. **[3]**

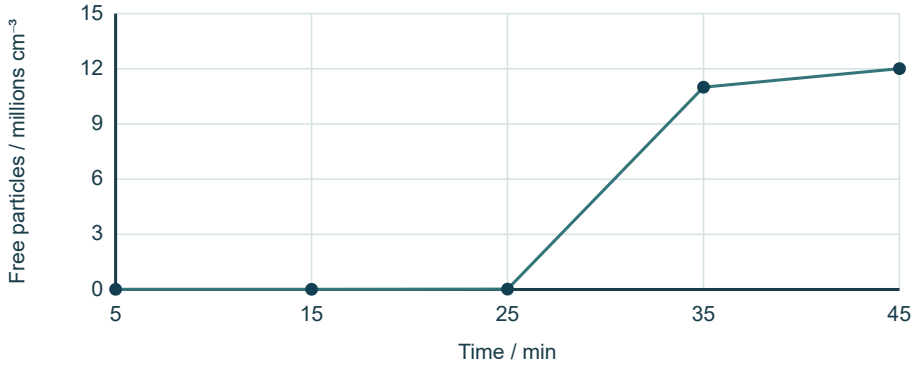
(d) Why would mean pressures be inadequate to determine the instant the aortic valve opens? State the necessary comparison and the consequence of valve closure. **[3]**

SECTION A · COMPULSORY
QUESTION 4 [12]

Following viral replication

A bacterial culture receives bacteriophages at time zero. After a short adsorption period the cells are washed. At each time, researchers assay free infective particles in a filtered sample. The experiment starts with 2.0×10^{5} infected bacteria per cm^{3}. Assume one lytic cycle and negligible decay.

Time / min	5	15	25	35	45
Free infective particles / cm^{3}	10000	10000	20000	11000000	12010000



Free particles / millions cm^{-3} plotted against Time / min. Points are measured means or observations as tabulated.

(a) Estimate the burst size from the final increase above residual free particles. State the necessary assumption about infected bacteria. **[3]**

(b) Explain why a low early free-particle count does not mean viral nucleic acid is absent. **[3]**

(c) Distinguish lytic replication from a lysogenic association using the fate of the host and viral genome. **[3]**

(d) Suggest how to test whether a resistant surviving bacterial colony carries a prophage and evaluate a positive DNA result. **[3]**

SECTION B · EXTENDED RESPONSE
QUESTION 5 [16]

From enzyme shape to metabolic control

A cell regulates a branched pathway. A product inhibits an enzyme near the beginning of its own branch.

(a) Outline the role of activation energy and how an enzyme changes reaction rate. **[3]**

(b) Explain how an enzyme–substrate interaction can generate specificity and catalysis. **[5]**

(c) Explain feedback inhibition and how a student could distinguish reversible competitive inhibition from a reduction in maximum catalytic capacity.

[1] Overall quality of the extended response.

SECTION B · EXTENDED RESPONSE
QUESTION 6 [16]

Water movement from soil to air

A terrestrial plant has access to moist soil while its leaves are exposed to dry air. Consider water potential and transport through the plant.

(a) Distinguish solute potential from pressure potential and relate them to total water potential in a cell. **[3]**

(b) Explain water uptake from soil into a root and onward delivery to xylem when the gradient is favourable. **[5]**

(c) Explain the cohesion–tension mechanism and how a prolonged drought can limit carbon gain. **[7]**

[1] Overall quality of the extended response.

SECTION B · EXTENDED RESPONSE
QUESTION 7 [16]

Chromosomes, linkage and variation

A diploid organism is heterozygous at two loci on the same chromosome. Its gametes show two common and two less common allele combinations.

(a) Outline how homologous chromosomes differ from sister chromatids. **[3]**

(b) Explain the events in meiosis that allow gametes to receive one allele at each locus and different combinations at linked loci. **[5]**

(c) Discuss how test-cross progeny can be used to infer linkage, with limits on the inference. **[7]**

[1] Overall quality of the extended response.